Experiment 7

Comparative Study of Popular No-Code / Low-Code Application Development Platforms

Aim

To study and compare some popular no-code and low-code application development platforms and to identify which platform is suitable for which type of requirement.

Objectives

- To understand the meaning of no-code and low-code development.
- To study the features of platforms such as Zoho Creator, Microsoft Power Apps, Google AppSheet, Bubble and OutSystems.
- To compare the platforms using common criteria.
- To suggest a suitable platform for different business needs.

Introduction

Traditional software development needs skilled programmers and a lot of time. No-code platforms allow people with no programming knowledge to build applications by using drag-and-drop tools, forms and ready-made templates. Low-code platforms are similar, but they also allow developers to write a small amount of code when more control is needed. These platforms help businesses to build applications faster and at lower cost.

Difference between No-Code and Low-Code

Point	No-Code	Low-Code
Target users	Business users, students, citizen developers	Developers and IT teams
Coding needed	None	Little coding for custom logic
Flexibility	Limited to what the platform offers	Higher, can be extended with code
Development speed	Very fast	Fast
Typical use	Forms, simple workflows, small apps	Larger business systems and integrations

Platforms Studied

- **Zoho Creator:** Low-code platform with forms, reports, pages, workflows and its own scripting language called Deluge. Part of the Zoho family of business apps.
- **Microsoft Power Apps:** Builds apps that connect easily to Excel, SharePoint, Teams and other Microsoft 365 tools.
- **Google AppSheet:** Creates mobile and web apps directly from Google Sheets, Excel or databases.
- **Bubble:** No-code tool mainly for building web applications with a visual editor and its own database.
- **OutSystems:** Enterprise low-code platform used for large and complex applications.

Comparison of the Platforms

Criteria	Zoho Creator	Power Apps	AppSheet	Bubble	OutSystems
Type	Low-code	Low-code	No-code	No-code	Low-code
Ease of learning	Easy	Medium	Very easy	Medium	Hard
Data source	Built-in database	Dataverse, Excel, SharePoint	Sheets, Excel, SQL	Built-in database	Own and external databases
Custom coding	Deluge script	Power Fx formulas	Expressions	Plugins	Full code extension
Mobile apps	Yes	Yes	Yes	Limited	Yes
Workflow / approval	Yes	Yes (Power Automate)	Yes	Yes	Yes
Pricing	Free trial, paid plans	Paid with Microsoft licences	Free for small use, paid plans	Free plan, paid plans	Costly, enterprise
Best suited for	Small and medium business apps	Organisations using Microsoft 365	Quick apps from spreadsheets	Startups and web products	Large enterprises

Procedure

1. Create or open free or trial accounts on the platforms listed above.
2. Build a simple "Student Record" form on each platform with the fields Name, Roll Number, Department and Email.
3. Note the time taken, the steps involved and the difficulty faced on each platform.
4. Check whether reports, mobile access and workflow options are available.
5. Collect the observations in a table and compare them.

Observations

Platform	Time to Build the Form	Difficulty (1 to 5)	Remarks
Zoho Creator	About 5 minutes	2	Simple form builder with ready-made field types
Power Apps	About 15 minutes	3	Needs a data source to be created first
AppSheet	About 7 minutes	2	App is created from a Google Sheet automatically
Bubble	About 20 minutes	4	Design and database must be set up separately

The values above are approximate and may change from person to person depending on experience.

Advantages and Limitations

Advantages	Limitations
Applications are built in days instead of months	Platform dependency; moving to another platform is difficult
Lower development cost and less need for programmers	Limited customisation in some no-code tools
Business users can make changes themselves	Monthly charges increase with more users and data
Built-in security, hosting and backup	Performance may reduce for very large applications

Selection Guide

- Choose **Zoho Creator** when forms, reports and approvals are needed at a reasonable cost.
- Choose **Power Apps** when the organisation already works with Microsoft 365.
- Choose **AppSheet** when the data is already in Google Sheets or Excel.
- Choose **OutSystems** for large, complex systems with a strong IT team.

Result

The selected no-code and low-code platforms were studied and compared on features, ease of use, cost and suitability. Each platform was found to be useful for a different kind of requirement.

Conclusion

No-code and low-code platforms reduce the time and effort needed to create business applications. Zoho Creator is a good choice for small and medium businesses that need forms, reports and approvals, Power Apps suits organisations that already use Microsoft tools, AppSheet is best for quick apps based on spreadsheets, and OutSystems is meant for large enterprise systems. The right platform should be selected by considering the budget, the skill of the team and the size of the application.